



Quantification of 11 kratom alkaloids including mitragynine and its main metabolites in human plasma using LC-MS/MS

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Received: 9 October 2024 / Revised: 27 November 2024 / Accepted: 29 November 2024 / Published online: 7 December 2024
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Abstract

Recently in the USA, kratom consumers increasingly report use of the plant for self-treatment of mood ailments, the lack of energy, chronic pain, and opioid withdrawal and dependence. Several alkaloids are present in kratom leaves, but limited data are available on their pharmacokinetics/pharmacodynamics, except for mitragynine. To support clinical studies, a high-performance liquid chromatography-tandem mass spectrometry assay for the simultaneous quantification of 11 kratom alkaloids in human plasma was developed and validated. For calibration standards and quality control samples, human plasma was fortified with alkaloids at varying concentrations, and 200 µL were extracted employing a simple one-step protein precipitation procedure. The extracts were analyzed using LC-MS/MS including electrospray ionization (ESI) in positive multiple reaction monitoring (MRM) mode. The lower limit of quantification was 0.5 ng/mL, and the upper limit of quantification was 400 ng/mL for all analytes. Inter-day analytical accuracy and imprecision ranged from 98.4 to 113% of nominal and from 3.9 to 14.7% (coefficient of variance), respectively. The analysis of plasma samples collected during a clinical trial administering capsules containing kratom leaf extract showed that most samples had quantifiable concentrations of mitragynine, 7-OH-mitragynine, speciogynine, speciociliatine, and paynantheine and that mitragynine, speciogynine, and speciociliatine accumulated in human plasma after daily administration over 15 days. An LC-MS/MS assay for the specific quantification of kratom alkaloids including mitragynine and its main metabolites was developed and successfully validated in human plasma. Human plasma samples collected following multiple oral administrations of a controlled Kratom extract documented accumulation of kratom alkaloids over 15 days.

Keywords Mitragynine · Kratom · LC-MS/MS · Pharmacokinetics

Introduction

Mitragyna speciosa Korth, also known as kratom, is a plant from the Rubiaceae (coffee) family indigenous to South East Asia [1–3]. Kratom has dose-dependent opioid- and psychostimulant-like properties [1, 4]. Said dose-dependent effects of kratom have primarily been attributed to mitragynine as well as its active metabolites 7-hydroxy-mitragynine

(7-OH-MTG) and mitragynine pseudo-indoxyl (MTG-pseudo indoxyl) [1, 3, 5]. Mitragynine, 7-OH-MTG, MTG-pseudo indoxyl, and 3-dehydro-mitragynine (3-dehydro-MTG) target a variety of receptors in the central nervous system including mu, delta, and kappa opioid receptors as well as serotonergic, adrenergic, dopaminergic, and adenosine receptors, albeit with significant differences in binding affinity [1–3, 6–8]. Reports on mitragynine's interaction with opioid receptors are inconsistent, with some suggesting partial agonism at the mu opioid receptor [3, 9] and others suggesting full agonist activity [10]. Overall, while the affinity of mitragynine for opioid receptors is less than that of morphine, 7-OH-MTG is more potent with approximately 46 times the affinity of mitragynine and 13 times that of morphine [5, 10] and MTG-pseudo indoxyl with approximately 100 times more than mitragynine and 20 times more than morphine [5]. *In vitro*, 3-dehydro-MTG shows binding

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affinity similar to 7-OH-MTG and partial agonist activation of the mu opioid receptor [6]. In recent years, other minor alkaloids present in the *Mitragyna speciosa* Korth plant were isolated such as corynantheidine, paynantheine, speciogynine, and speciociliatine [11]. Their chemical structures are closely related to mitragynine. *In vitro* they show low-potency, competitive antagonism of the opioid receptors, but *in vivo* their effects suggest serotonin receptor activation [3, 11–13].

Kratom is usually consumed orally by chewing fresh or dried leaves, drinking crushed or boiled leaves in a tea, or by consuming encapsulated leaf powder [4]. Kratom leaves are traditionally taken by indigenous peoples to treat common illnesses and are popular for their energizing and pain alleviating effects [3, 14]. In recent years, kratom also gained popularity in the USA for self-treatment of chronic pain, anxiety, opioid withdrawal, and dependence [14–17]. An estimated 2.1 million US residents used kratom in 2020 [18]. Due to its growing popularity, the US National Poison Data System reported a 40-fold increase in cases of non-lethal adverse reactions associated with kratom use between 2010 and 2019, despite the low rate of adverse effects reported in recent surveys [15, 17, 18]. The most common adverse reactions were mild such as irritability and tachycardia, while life-threatening conditions were rare [15, 17, 18]. Kratom tolerance and opioid-like withdrawal syndromes have been reported among chronic frequent users and their likelihood and severity depend on duration, frequency, and intensity of kratom use [15, 17, 18]. The legal status of kratom varies across countries and within states across the USA from legal to illegal mainly due to concerns for its abuse potential [17]. Several kratom-associated deaths have been reported, but, in most cases, kratom was listed only as a contributing factor due to the presence of other psychoactive substances [16–18]. Overall, the pharmacokinetics (PK)/pharmacodynamics (PD) and toxicokinetics/toxicodynamics of kratom alkaloids in humans remain poorly understood [3, 14].

As of today, several methods for the screening and quantification of mitragynine in different biological matrices, including urine, postmortem blood, umbilical cord, and rat plasma, have been published [19–29]. Most assays employ liquid-liquid extraction (LLE) or solid phase extraction (SPE) protocols combined with liquid chromatography-tandem mass spectrometry (LC-MS/MS) [19, 25, 26]. Few analytical methods included the major mitragynine metabolite 7-OH-MTG or other kratom alkaloids [19, 22, 24, 30–36]. No validated assay for the comprehensive quantification of mitragynine, its three major metabolites and seven major kratom alkaloids in human plasma have been published yet.

Here, we report the development and validation of such an LC-MS/MS assay combined with a one-step protein precipitation extraction procedure for the simultaneous quantification of 11 kratom alkaloids including mitragynine and

its main metabolites. After successful validation, as proof of concept, plasma study samples from a clinical trial assessing kratom alkaloids pharmacokinetics after oral administration of kratom leaf extract to healthy participants were analyzed with the present assay.

Materials and methods

Chemicals and reagents

Mitragynine, 7-hydroxy-mitragynine (7-OH-MTG), speciociliatine, speciogynine, paynantheine, corynantheidine, mitraphylline, corynoxine, and iso-corynoxine were purchased from Cayman Chemical (Ann Arbor, MI, USA). The reference material of mitragynine pseudo-indoxyl (MTG-pseudo indoxyl) and 3-dehydro-mitragynine (3-dehydro-MTG) were obtained from Jiangyin PharmaAdvance (Jiangyin, Jiangsu Province, PR China). The deuterated internal standard compounds mitragynine-d3 (MTG-d3) and 7-hydroxy-mitragynine-d3 (7-OH-MTG-d3) were acquired from Cayman Chemicals. All reference compounds had a valid certificate of analysis. Chemical structures and International Union of Pure and Applied Chemistry (IUPAC) names are presented in Figure S1 and Table S1.

The chemicals and solvents for this study including methanol, acetonitrile, water, and formic acid (all LC/MS grade) were purchased from Fisher Scientific (Waltham, MA, USA).

Blank human K₂EDTA plasma was from BioIVT (Westbury, NY, USA).

Instrumentation

High-performance liquid chromatography- tandem mass spectrometry (LC-MS/MS) analysis was performed on a Sciex API5000 triple quadrupole mass spectrometer with a Turbo V Ion Spray source (Sciex, Concord, ON, Canada). The high-performance liquid chromatography (HPLC) system consisted of an Agilent 1260 (G1312B) binary pump, a G1379B vacuum degasser, and an Agilent 1260 (G1316C) thermostated column compartment (all Agilent Technologies, Palo Alto, CA, USA). The system also contained an HTC-xt Leap Autosampler (Leap Technologies, Carrboro, NC, USA). Analyst 1.6.2 was employed for data acquisition and Sciex OS version 1.6.1 for data analysis (Sciex, Concord, ON, Canada).

Calibrators, quality control (QC), and internal standards

Stock solutions were purchased at a concentration of 1 mg/mL in methanol. Separate working calibrator and quality

control stock solutions were prepared in methanol. Calibrators contained 0.25, 0.5, 1, 2.5, 5, 10, 25, 50, 100, 200, and 400 ng/mL of all compounds. QC working solutions were prepared yielding QC samples at the lower limit of quantification (LLOQ), 3xLLOQ, 15, 120, and 320 ng/mL for all compounds. Calibrators and QC samples were obtained by adding 20 μ L working solution to 180 μ L blank human plasma. The internal standard stock solution was added to the protein precipitation solution (acetonitrile) at a final concentration of 1 ng/mL for MTG-d3 and 2.5 ng/mL for 7-OH-MTG-d3. The protein precipitation/internal standard working solution was prepared freshly before each extraction. All methanolic working standard solutions were stored in amber glass vials at -80°C .

Sample preparation

Calibrators and quality control samples were fortified each day of analysis. K_2EDTA plasma aliquots of 200 μ L of calibrators, quality controls, or blank samples were transferred into 1.5 mL low binding polypropylene vials with conical bottoms and snap-on lids. Eight hundred microliter acetonitrile containing the internal standards were added. Samples were vortexed for 10 min and centrifuged (25,000 $\times$ g, 4°C for 10 min, MR 23i centrifuge with a Fiberlite rotor, Thermo Scientific, Waltham, MA). The supernatant was transferred into a glass HPLC vial and placed into the temperature-controlled autosampler ($+4^{\circ}\text{C}$).

LC-MS/MS

Four (4) μ L of the sample extracts were injected onto a 2.7 μ m, 4.6×150 mm Poroshell 120 EC-C18 (Agilent Technologies, Palo Alto, CA). Analytes were eluted from the analytical column using a gradient. Mobile phase A was 0.1% formic acid in HPLC grade water, and B was 0.1% formic acid in methanol at a flow rate of 0.8 mL/min. The gradient program was 30% B for 0.5 min to 50% B at 0.7 min, held for 1.3 min, increased to 80% B at 6 min and to 100% B at 6.5 min, held for 0.5 min, re-equilibrated to starting conditions (30% B) over 0.2 min, and held for 0.8 min. Total run time was 8 min and the analytical column was kept at 60°C . MS/MS data were acquired after electrospray ionization (ESI) in combination with multiple reaction monitoring (MRM) in the positive ion mode. ESI source parameters were curtain gas flow 40 L/min, CAD gas flow 9 L/min, source temperature 600°C , ion spray voltage 5500V, and ion source gas 1 and 2 80 L/min. Individual MS/MS settings are listed in Table S2, and Figure S1 depicts the proposed MS/MS fragmentation pattern based on the work from Avula et al. [37].

Data analysis

Calibration curves were constructed for each batch from analyte/internal standard peak area ratios. Calibrators were fitted using a quadratic equation in combination with $1/x$ weighting for all analytes, except 7-OH-MTG, MTG-pseudoindoxyl, and corynantheidine that were fitted using a linear equation in combination with $1/x$ weighting. Calculations were carried out using Sciex MultiQuant OS (version 1.6.1).

Method validation

Method development and validation were conducted following applicable Clinical Laboratory Standards Institute (CLSI) and Food and Drug Administration (FDA) guidelines for bioanalytical assays [38–40]. For details, please see the Supplementary Materials.

Proof of concept for analysis of plasma samples collected during a single ascending/multiple ascending dose clinical trial

A subset of plasma samples collected as part of a double-blind, placebo-controlled clinical trial investigating the pharmacokinetics, safety, and tolerability of kratom formulations in healthy participants was analyzed [41]. A total of 365 plasma samples were obtained from 10 healthy participants enrolled in the clinical trial and randomized to single and repeated oral doses of kratom leaf extract capsules administered once daily. Third party analytical testing (Cambium Analytica, Traverse City, MI, USA) confirmed that each capsule contained 25 mg *Mitragynine speciosa* extract including 9.87 mg MTG, 0.91 mg speciogynine, 1.86 mg speciociliatine, 0.77 mg paynantheine, and 0.98 mg corynoxine. Each participant received single and multiple doses of kratom corresponding to a daily dose of 9.87 mg, 29.61 mg, and 59.22 mg of MTG for the low-, mid-, and high-dose group, respectively. After the single dose administration, a 10-day wash-out period was observed before starting the 15-day multiple dose administration. Informed written consent was obtained from all participants. The study was conducted in accordance with the Declaration of Helsinki including its amendments and was approved by the Advarra Institutional Review Board (Columbia, MD, USA, Pro00048457).

The pharmacokinetics of the kratom alkaloids was performed by non-compartmental analysis using Phoenix WinNonlin, version 8.3.5. (Certara, Princeton, NJ, USA). Concentrations below the LLOQ but above the lower limit of detection (LLOD, 0.1 ng/mL for all compounds) were estimated by extrapolation of the calibration curve below the lowest calibrator and said concentrations were included in the analysis. The rationale was that this constituted less of an error than setting concentrations below the LLOQ to 0

ng/mL if this was clearly not the case. Concentrations below the LLOD were assigned a concentration of 0 ng/mL for the pharmacokinetics and distribution statistics analysis.

Results

Several columns with different stationary phase materials were tested with the goal of chromatographically separating the kratom alkaloids. The Agilent Poroshell 120 EC-C18 analytical column achieved baseline separation for all analytes (Figure S2). Representative extracted ion chromatograms of a blank and a blank sample spiked with analyte concentrations at the LLOQ are shown in Figures S2 and Fig. 1 for a participant sample.

The results of the assay validation can only briefly be summarized here. For details, please see the Supplemental Materials. Eight plasma samples collected from eight different individuals who did not receive kratom showed no

endogenous interfering peaks for analytes exceeding 20% of the corresponding analyte signals at the LLOQ. The LLOQ was 0.5 ng/mL, and the upper limit of quantification was 400 ng/mL for all analytes (for more details, please see Table 1). Several fits for calibration curves for the various analytes were assessed. A linear fit in combination with $1/x$ weighting consistently gave the best results for 7-OH-MTG, MTG-pseudo indoxyl, and corynantheidine, while a quadratic fit in combination with $1/x$ weighting was best for all the other analytes. All calibration curve correlation coefficients (r) and coefficients of determination (r^2) exceeded 0.99 (Table S3). Analytical recovery and imprecision testing results are shown in summary in Table 2 and in detail in Tables S4 and S5. Intra-day analytical accuracies and imprecisions ranged from 102 to 113% of the nominal concentrations and from 1.4 to 8.5% (coefficient of variance (CV)), respectively, and inter-day analytical accuracies from 98.4 to 113% of the nominal concentrations and imprecisions from 3.9 to 14.7% CV. Mean absolute recoveries were between

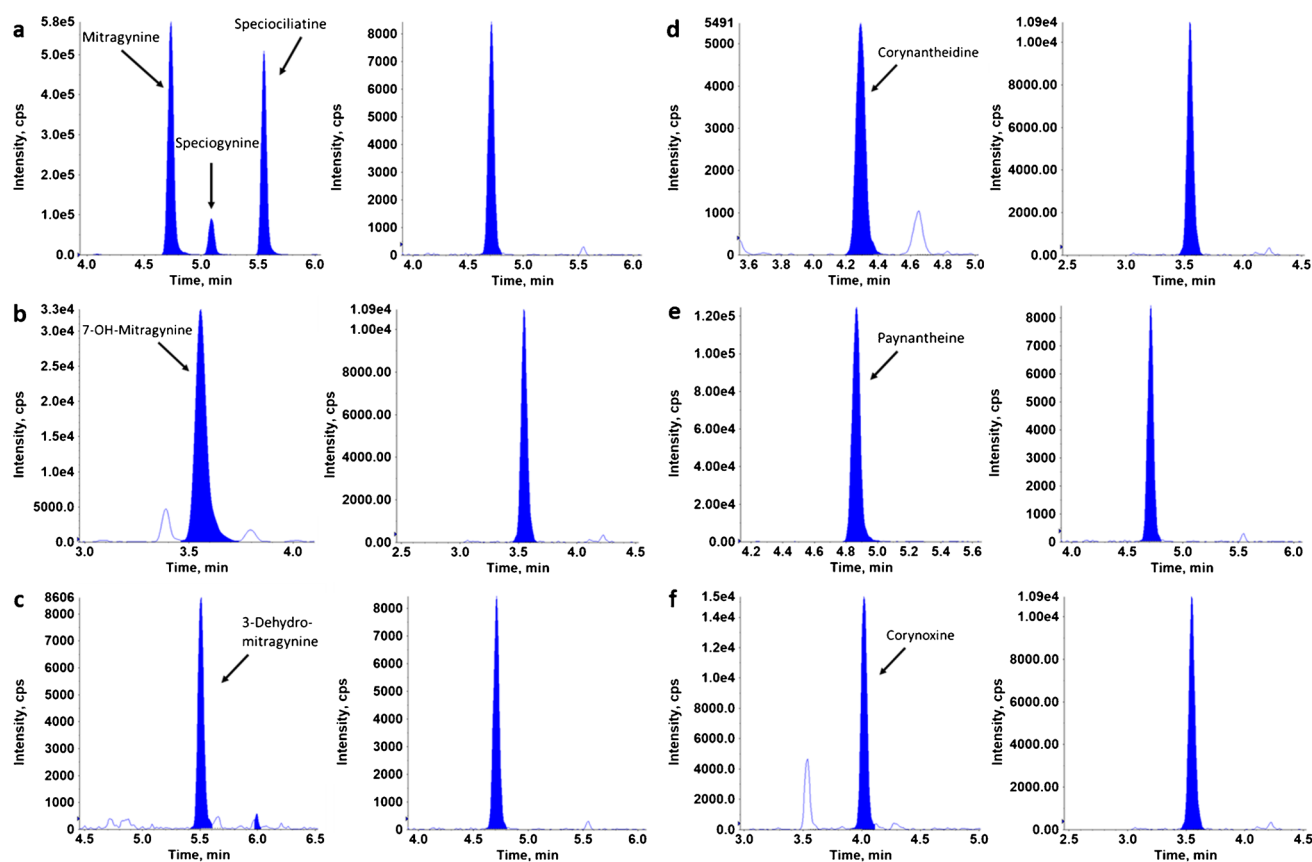


Fig. 1 Representative extracted ion chromatograms of plasma samples from a participant after oral administration of kratom leaves extract. The left extracted ion chromatogram shows the analytes of interest, the right shows the corresponding internal standard (IS). The concentrations for this sample were **a** mitragynine 229 ng/mL, speciogynine 25.2 ng/mL, speciociliatine 151 ng/mL, IS mitragynine-d3;

b 7-hydroxy-mitragynine 39.4 ng/mL, IS 7-hydroxy-mitragynine-d3; **c** 3-dehydro-mitragynine 3.01 ng/mL, IS mitragynine-d3; **d** corynantheidine 3.29 ng/mL, IS 7-hydroxy-mitragynine-d3; **e** paynantheine 32.8 ng/mL, IS mitragynine-d3; **f** corynoxine 3.63 ng/mL, IS 7-hydroxy-mitragynine-d3

Table 1 Concentration range, mean recovery, absolute and relative matrix effects for kratom alkaloids in human plasma. *MTG* mitragynine, *Conc.* concentration; *std* standard deviation

Analyte	Internal standard	Conc. range (ng/mL)	Recovery Mean ($\pm$ std)	Absolute matrix effect Mean ($\pm$ std)	Relative matrix effect Mean ($\pm$ std)
Mitragynine	MTG-d3	0.5–400	91.6 ($\pm$ 7)	8.9 ($\pm$ 4.5)	–3.1 ($\pm$ 4.4)
7-OH-MTG	7-OH-MTG-d3	0.5–400	99.8 ($\pm$ 10.6)	3 ($\pm$ 3.6)	–36.9 ($\pm$ 2.9)
3-Dehydro-MTG	MTG-d3	0.5–400	89.7 ($\pm$ 9.1)	59.4 ($\pm$ 28.3)	41.5 ($\pm$ 22.8)
MTG-pseudo indoxyl	7-OH-MTG-d3	0.5–400	101.3 ($\pm$ 11.1)	–1.6 ($\pm$ 1.6)	–14.3 ($\pm$ 4.7)
Speciogynine	MTG-d3	0.5–400	91.1 ($\pm$ 6.9)	10.8 ($\pm$ 4.2)	–1.3 ($\pm$ 4.6)
Speciociliatine	MTG-d3	0.5–400	91.3 ($\pm$ 8.1)	14.3 ($\pm$ 4.6)	1.8 ($\pm$ 4.8)
Corynantheidine	7-OH-MTG-d3	0.5–400	97.6 ($\pm$ 8.2)	6.8 ($\pm$ 1.3)	–7.1 ($\pm$ 3.4)
Paynantheine	MTG-d3	0.5–400	91.3 ($\pm$ 6.1)	14.1 ($\pm$ 4.3)	1.5 ($\pm$ 4.4)
Mitraphylline	MTG-d3	0.5–400	89.4 ($\pm$ 6.7)	34.3 ($\pm$ 4.2)	19.7 ($\pm$ 7.6)
Corynoxine	7-OH-MTG-d3	0.5–400	95.3 ($\pm$ 7.1)	61.9 ($\pm$ 5.9)	40.6 ($\pm$ 3.1)
Iso-Corynoxine	7-OH-MTG-d3	0.5–400	97.1 ($\pm$ 6.9)	–1.5 ($\pm$ 2)	–14.3 ($\pm$ 2.6)

Table 2 Summary of analytical bias and imprecision for kratom alkaloids in human plasma. *MTG* mitragynine, *std* standard deviation, *%CV* % coefficient of variation, *%nominal* % of nominal concentration

Analyte	Intra-day (<i>n</i> =6/level)		Inter-day (<i>n</i> =36/level)	
	Accuracy	Imprecision	Accuracy	Imprecision
	%nominal ($\pm$std)	% CV	%nominal ($\pm$std)	% CV
MTG	107 ($\pm$ 2.5)	2.3	107 ($\pm$ 4.8)	4.5
7-OH-MTG	107 ($\pm$ 3.4)	3.2	102 ($\pm$ 3.0)	3.0
3-Dehydro-MTG	107 ($\pm$ 1.3)	1.2	106 ($\pm$ 2.8)	2.6
MTG-pseudo indoxyl	108 ($\pm$ 4.1)	3.8	107 ($\pm$ 2.8)	2.6
Speciogynine	107 ($\pm$ 3.8)	3.6	106 ($\pm$ 3.9)	3.7
Speciociliatine	105 ($\pm$ 2.8)	2.6	106 ($\pm$ 3.4)	3.2
Corynantheidine	108 ($\pm$ 3.3)	3.0	106 ($\pm$ 3.4)	3.2
Paynantheine	108 ($\pm$ 2.7)	2.5	106 ($\pm$ 3.5)	3.3
Mitraphylline	106 ($\pm$ 2.1)	1.9	106 ($\pm$ 3.5)	3.3
Corynoxine	105 ($\pm$ 1.3)	1.2	104 ($\pm$ 2.0)	1.9
Iso-Corynoxine	105 ($\pm$ 2.0)	1.9	103 ($\pm$ 1.8)	1.8

89.4 $\pm$ 6.7% for mitraphylline and 101.3 $\pm$ 11.1% for MTG-pseudo indoxyl. Absolute matrix effects were between –1.6 $\pm$ 1.6% for MTG-pseudo indoxyl and 61.9 $\pm$ 5.9% for corynoxine. Relative matrix effects were between –36.9 $\pm$ 2.9% for 7-OH-MTG and 41.5 $\pm$ 22.8% for 3-dehydro-MTG. Recoveries and absolute and relative matrix effects are shown in detail in Table 1. All analytes were stable when stored at room temperature for up to 4 h with concentrations within $\pm$ 7.9% of the nominal concentrations and refrigerated

up to 1 week and 1 month at 4°C with concentrations within $\pm$ 4.4% and 11.2%, respectively, except for 7-OH-MTG ($\pm$ 27.8% and $\pm$ 20.2%) and MTG-pseudo indoxyl ($\pm$ 35.8% and $\pm$ 35.1%). All analytes were stable when stored up to 6 months at –20°C with concentrations within $\pm$ 14.7% of the nominal concentrations, except for 7-OH-MTG ($\pm$ 24.3%) and MTG-pseudo indoxyl ($\pm$ 28.4%). All analytes were stable when stored up to 6 months at –80°C with concentrations within $\pm$ 15.1% of the nominal concentrations. After

three freeze-thaw cycles (-80°C), concentrations for all analytes were within $\pm 10.9\%$ of the nominal concentrations. Stability in the autosampler at $+4^{\circ}\text{C}$ was within $\pm 10.4\%$ of the nominal concentrations for all analytes after 72 h, except for corynoxine with $\pm 23.0\%$. All stability testing results are shown in detail in Tables S6–S8 in the Supplementary Materials. No significant carryover was observed for any of the kratom alkaloids and MTG metabolites after injecting blank extracted plasma samples after the highest calibrator. The accuracies of the diluted samples (1:5, 1:10, 1:20, and 1:50) were within 85–115% for all compounds except for MTG-pseudo indoxyl when diluted 1:50 and iso-corynoxine when diluted 1:5 and 1:50; none of the dilutions were successful for corynoxine (Table S9).

Of the ten participants included in the single ascending/multiple ascending dose pharmacokinetics study of kratom leaf extract capsules after oral administration, one received the low dose of kratom, three the mid dose, and six the high dose (a daily dose of 9.87 mg, 29.61 mg, and 59.22 mg MTG, respectively). One participant in the high-dose group dropped out of the study during the single dose session and was not included in the analysis. Two participants in the same group completed the single dose administration but dropped out during the multiple dose phase.

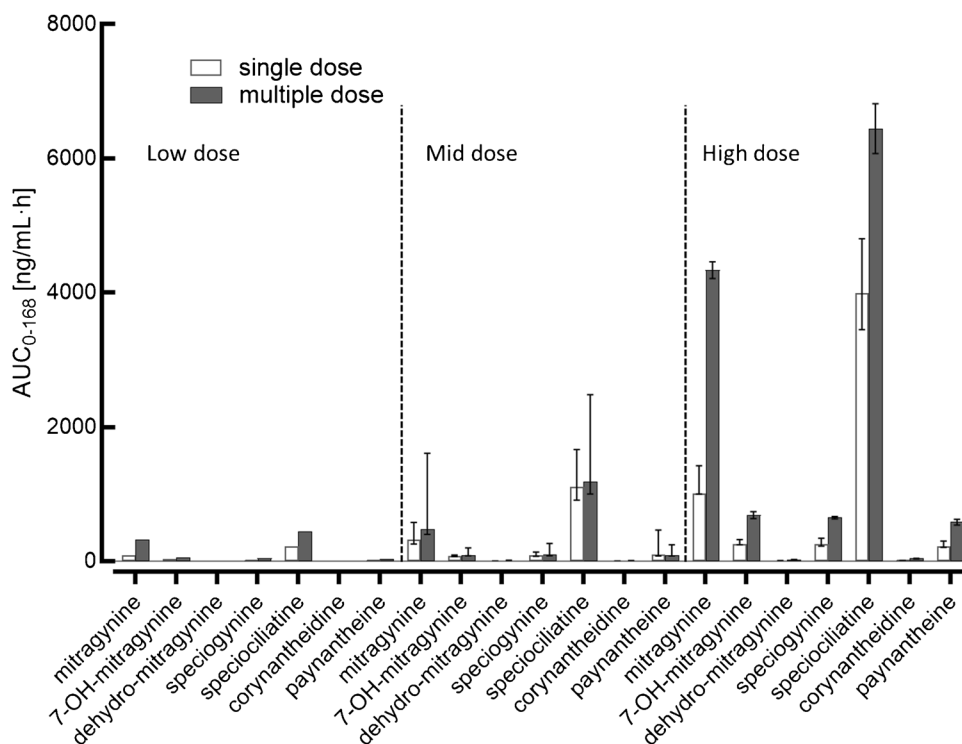
Pharmacokinetic parameters were calculated for all analytes and doses with sufficient positive results and are summarized in Table S10 and Figs. 2 and S3–S8. Quantifiable concentrations of MTG, 7-OH-MTG, speciogynine, speciociliatine, and paynantheine were detected in most samples

in the different dosing groups, while 3-dehydro-MTG and corynantheidine were not detectable in the low-dose group, and corynoxine was detected only in the high-dose group. MTG-pseudo indoxyl, mitraphylline, and iso-corynoxine were not detected in any of the plasma samples. Comparison of median $\text{AUC}_{0-168\text{h}}$ after single and multiple doses (on day 15) showed systemic accumulation for mitragynine and speciociliatine for all dose groups (Table S10). Mitragynine had an accumulation of 3.5-, 1.5-, and 4.3-fold and speciociliatine of 2-, 1.1-, and 1.4-fold for the low, mid, and high group, respectively. At the high-dose, 7-OH-mitragynine, 3-dehydro-MTG, speciogynine, and paynantheine showed an accumulation of 2.7-, 7.3-, 2.6-, and 2.6-fold, respectively (Figs. 2 and S7–S8). Each capsule of 25 mg *Mitragynine speciosa* extract contains 9.87 mg mitragynine and only 1.86 mg speciociliatine. However, speciociliatine achieved median $\text{AUC}_{0-168\text{h}}$ and C_{max} values higher than mitragynine dosing groups suggesting higher relative bioavailability (Table S10 and Figs. 2 and S7).

Discussion

Interest in potential therapeutic applications of kratom leaves and extracts significantly increased in recent years, but little is known about the pharmacokinetics/pharmacodynamics of kratom alkaloids other than mitragynine [1, 3, 4, 26]. For these reasons, we developed and validated an LC-MS/MS assay for the screening and quantification of 11 alkaloids

Fig. 2 Comparison of the areas-under-time-concentration curves over 168 h ($\text{AUC}_{0-168\text{h}}$) after single and multiple (over 15 days) oral administrations of kratom leaf extract capsules at three different doses. For mitragynine (low dose, 9.87 mg; mid dose, 29.6 mg; high dose, 59.2 mg), 7-hydroxy-mitragynine, 3-dehydro-mitragynine, speciogynine (low dose, 0.91 mg; mid dose, 2.73 mg; high dose, 5.46 mg), speciociliatine (low dose, 1.86 mg; mid dose, 5.58 mg; high dose, 11.2 mg), corynantheidine and paynantheine (low dose, 0.77 mg; mid dose, 2.31 mg; high dose, 4.62 mg)



including MTG and its main metabolites in human plasma. The present assay was applied to the analysis of samples collected during a double-blind, randomized, placebo-controlled clinical trial [41].

The assay achieved sensitivity for all compounds comparable to previously published methods using only 0.2 mL plasma and a simple protein precipitation extraction procedure [20, 28, 36]. The present assay fully separates MTG and its diastereoisomers as well as multiple hydroxylated metabolites. Tanna et al. quantified seven kratom alkaloids in 0.2 mL human plasma, achieved LLOQs similar to the present method, but provided only limited validation results [36]. Hughs et al. included MTG, 7-OH-MTG, speciociliatine, speciogynine, and mitraciliatine in an assay to quantify said compounds in umbilical cord tissue [22], and Basiliere and Kerrigan quantified MTG, speciociliatine, speciogynine, paynantheine, and four MTG metabolites in human urine with LLOQs similar to the present method [19]. Few methods were developed for the quantification of specific kratom alkaloids. King et al. developed an LC-MS/MS method for the quantification of corynantheidine in rat plasma [30], while Philipp et al. studied the metabolism of speciogynine, speciociliatine, mitraciliatine, isopaynantheine, and paynantheine in rat and human urine [31–35]. We developed and validated the present assay in human plasma to support ongoing clinical trials including regulatory clinical trials.

The matrix effect experiments showed acceptable matrix effects (ion suppression or enhancement) for all compounds [39]. All kratom alkaloids were stable at room temperature for up to 4 h and when stored up to 6 months at -80°C , while under the other tested conditions, 7-OH-MTG seems to convert to MTG-pseudo indoxyl as reported by Kamble et al. [5].

Overall, the assay proved robust and suitable for the analysis of large sets of clinical study samples. Here, as a representative example, we report the analysis of 365 human EDTA plasma samples from a double-blind, placebo-controlled clinical trial following oral single and multiple kratom leaf extract capsules. Overall, speciociliatine showed better bioavailability compared to mitragynine. Speciociliatine administered doses were $\sim 1/5$ those of mitragynine; still speciociliatine achieved higher C_{max} and AUCs for all conditions tested. Based on the median $\text{AUC}_{0-168\text{h}}$ values after single and multiple doses on study day 15, the relative oral bioavailability of speciociliatine in the highest dose group was 7.9-fold higher than that of mitragynine. There are no human studies on the oral bioavailability of neither mitragynine nor speciociliatine. Nevertheless, our observations for mitragynine align with the published literature in rats that report oral bioavailability to be low ($\sim 21\%$) [9, 21], while oral bioavailability of speciociliatine in rats seems to be much lower ($\sim 21\%$) than in human [42]. Moreover, our assay allowed first insights into the systemic accumulation

of kratom alkaloids after repeated oral doses of kratom leaf extract. Mitragynine (4.3-fold based on multiple dose/ single dose $\text{AUC}_{0-168\text{h}}$ in the highest dose group) and speciociliatine (1.4-fold in the highest dose group) showed accumulation at all doses, while 7-OH-mitragynine, 3-dehydro-mitragynine, speciogynine, and paynantheine only after oral administration of the highest doses of kratom leaf extract. MTG-pseudo indoxyl was not detected in the present study samples. These data support the importance of monitoring even minor kratom alkaloids that could contribute to the overall effects of kratom; for instance, speciociliatine showed higher oral bioavailability and systematic accumulation that potentially may affect the duration and intensity of its serotonergic effects. Finally, 7-OH-mitragynine achieved C_{max} and AUCs 4–5-fold lower than mitragynine. However, the high potency of 7-OH-mitragynine (46-fold that of mitragynine and 13-fold of morphine) would significantly contribute to the pharmacodynamic/toxicodynamic effects after kratom administration highlighting the importance of assessing metabolite concentrations in animal and clinical studies assessing the treatment effects and safety of kratom extracts.

To our knowledge, only Tanna et al. reported on the pharmacokinetics of different kratom alkaloids after oral administration of a single dose of kratom leaf tea [36]. The alkaloid composition of the kratom leaf used by Tanna et al. differs from the one used in the present study; the mitragynine dose is comparable to the mid dose of the present study. The speciogynine and speciociliatine doses in the study by Tanna et al. are comparable to the high dose in the present study, and the paynantheine dose is higher than any dose in the present protocol [36]. Mitragynine bioavailability seems to be comparable across both studies; however, apparent volume of distribution (V_z/F) and oral clearance (CL/F) were 4 and 2.5 times lower, which may be explained by a difference in oral bioavailability due to the difference in formulation. Overall, the present pharmacokinetic study provides proof of concept that the present assay covers the concentration ranges for the key kratom alkaloids and major mitragynine metabolites after oral administration of kratom leaf extract.

Conclusions

The present LC-MS/MS assay simultaneously quantifies 11 kratom alkaloids and metabolites in human plasma. The assay met predefined validation acceptance criteria and achieved sensitivity and performance suitable to support clinical diagnostics and exploratory and regulatory clinical trials. The pharmacokinetics of 11 analytes after oral administration of kratom leaf extract in healthy participants are described after analysis of clinical samples in a controlled pharmacokinetics study. The significant systemic exposure

to highly active mitragynine metabolites found in the present study emphasizes the importance of monitoring their formation in clinical studies to better understand kratom alkaloid pharmacokinetics/pharmacodynamics.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00216-024-05689-9>.

Acknowledgements The University of Colorado is a Certara Center of Excellence. The Center of Excellence program supports leading institutions with Certara's state-of-the-art, model-informed drug development software.

Author contribution Cristina Sempio, Jorge Campos-Palomino, Jelena Klawitter, Wanzhu Zhao, Uwe Christians, and Jost Klawitter developed and validated the analytical method including data analysis and interpretation. Marilyn Huestis conceived and executed the clinical study in collaboration with Johnson Food, LLC. Cristina Sempio drafted the article, and Uwe Christians, Jost Klawitter, and Marilyn Huestis edited the manuscript. All authors provided critical revisions to the article and approved the final version.

Funding Assay development and validation was funded by iC42 Clinical Research and Development, Department of Anesthesiology, University of Colorado, Aurora, Colorado, USA. The clinical study was funded by Johnson Foods, LLC (Sheridan, WY, USA).

Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval This clinical study was conducted in accordance with consensus ethics principles, International Conference on Harmonization Good Clinical Practice guidelines, the Declaration of Helsinki, its amendments, and local laws and regulations. The protocol was approved by the Advarra Institutional Review Board (Columbia, MD, USA, Pro00048457).

Consent to participate Written informed consent was obtained from each participant before any trial-related procedures were performed.

Conflict of interest Cristina Sempio, Jorge Campos-Palomino, Jelena Klawitter, Wanzhu Zhao, Uwe Christians, and Jost Klawitter are employees of the University of Colorado and do not have any conflicts or competing interests. Marilyn Huestis is a consultant to Johnson Foods who conducted the controlled kratom administration study and a consultant for the American Kratom Association, which had no role in this study or publication.

References

- Eastlack SC, Cornett EM, Kaye AD. Kratom-pharmacology, clinical implications, and outlook: a comprehensive review. *Pain Ther*. 2020;9(1):55–69. <https://doi.org/10.1007/s40122-020-00151-x>.
- Farkas DJ, Inan S, Heydari LN, Johnson CT, Zhao P, Bradshaw HB, et al. Cannabinoid mechanisms contribute to the therapeutic efficacy of the kratom alkaloid mitragynine against neuropathic, but not inflammatory pain. *Life Sci*. 2023;328:121878. <https://doi.org/10.1016/j.lfs.2023.121878>.
- Hiranita T, Obeng S, Sharma A, Wilkerson JL, McCurdy CR, McMahon LR. In vitro and in vivo pharmacology of kratom. *Adv Pharmacol*. 2022;93:35–76. <https://doi.org/10.1016/bbs.apha.2021.10.001>.
- Hassan Z, Muzaimi M, Navaratnam V, Yusoff NH, Suhaimi FW, Vadivelu R, et al. From Kratom to mitragynine and its derivatives: physiological and behavioural effects related to use, abuse, and addiction. *Neurosci Biobehav Rev*. 2013;37(2):138–51. <https://doi.org/10.1016/j.neubiorev.2012.11.012>.
- Kamble SH, Leon F, King TI, Berthold EC, Lopera-Londono C, Siva Rama Raju K, et al. metabolism of a kratom alkaloid metabolite in human plasma increases its opioid potency and efficacy. *ACS Pharmacol Transl Sci*. 2020;3(6):1063–8. <https://doi.org/10.1021/acscptsci.0c00075>.
- Chakraborty S, Uprety R, Slocum ST, Irie T, Le Rouzic V, Li X, et al. Oxidative metabolism as a modulator of kratom's biological actions. *J Med Chem*. 2021;64(22):16553–72. <https://doi.org/10.1021/acs.jmedchem.1c01111>.
- Kruegel AC, Grundmann O. The medicinal chemistry and neuropharmacology of kratom: a preliminary discussion of a promising medicinal plant and analysis of its potential for abuse. *Neuropharmacology*. 2018;134(Pt A):108–20. <https://doi.org/10.1016/j.neuropharm.2017.08.026>.
- Matsumoto K, Horie S, Ishikawa H, Takayama H, Aimi N, Ponglux D, et al. Antinociceptive effect of 7-hydroxymitragynine in mice: discovery of an orally active opioid analgesic from the Thai medicinal herb *Mitragyna speciosa*. *Life Sci*. 2004;74(17):2143–55. <https://doi.org/10.1016/j.lfs.2003.09.054>.
- Ya K, Tangamornsuksan W, Scholfield CN, Methaneethorn J, Lohitnavy M. Pharmacokinetics of mitragynine, a major analgesic alkaloid in kratom (*Mitragyna speciosa*): a systematic review. *Asian J Psychiatr*. 2019;43:73–82. <https://doi.org/10.1016/j.ajp.2019.05.016>.
- Yamamoto LT, Horie S, Takayama H, Aimi N, Sakai S, Yano S, et al. Opioid receptor agonistic characteristics of mitragynine pseudoinoxyl in comparison with mitragynine derived from Thai medicinal plant *Mitragyna speciosa*. *Gen Pharmacol*. 1999;33(1):73–81. [https://doi.org/10.1016/s0306-3623\(98\)00265-1](https://doi.org/10.1016/s0306-3623(98)00265-1).
- Karunakaran T, Ngew KZ, Zailan AAD, Mian Jong VY, Abu Bakar MH. the chemical and pharmacological properties of mitragynine and its diastereomers: an insight review. *Front Pharmacol*. 2022;13:805986. <https://doi.org/10.3389/fphar.2022.805986>.
- Kamble SH, Berthold EC, Kanumuri SRR, King TI, Kuntz MA, Leon F, et al. Metabolism of speciociliatine, an overlooked kratom alkaloid for its potential pharmacological effects. *AAPS J*. 2022;24(5):86. <https://doi.org/10.1208/s12248-022-00736-8>.
- Leon F, Obeng S, Mottinelli M, Chen Y, King TI, Berthold EC, et al. Activity of *Mitragyna speciosa* ("kratom") alkaloids at serotonin receptors. *J Med Chem*. 2021;64(18):13510–23. <https://doi.org/10.1021/acs.jmedchem.1c00726>.
- Hartley C 2nd, Bulloch M, Penzak SR. Clinical pharmacology of the dietary supplement kratom (*Mitragyna speciosa*). *J Clin Pharmacol*. 2022;62(5):577–93. <https://doi.org/10.1002/jcph.2001>.
- Grundmann O. Patterns of Kratom use and health impact in the US-Results from an online survey. *Drug Alcohol Depend*. 2017;176:63–70. <https://doi.org/10.1016/j.drugalcdep.2017.03.007>.
- Grundmann O, Hendrickson RG, Greenberg MI. Kratom: history, pharmacology, current user trends, adverse health effects and potential benefits. *Dis Mon*. 2023;69(6):101442. <https://doi.org/10.1016/j.disamonth.2022.101442>.
- Swogger MT, Smith KE, Garcia-Romeu A, Grundmann O, Veltri CA, Henningfield JE, et al. Understanding kratom use: a guide for

- healthcare providers. *Front Pharmacol.* 2022;13:801855. <https://doi.org/10.3389/fphar.2022.801855>.
18. Gorelick DA. Kratom: substance of abuse or therapeutic plant? *Psychiatr Clin North Am.* 2022;45(3):415–30. <https://doi.org/10.1016/j.psc.2022.04.002>.
 19. Basiliere S, Bryand K, Kerrigan S. Identification of five Mitragyna alkaloids in urine using liquid chromatography-quadrupole/time of flight mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2018;1080:11–9. <https://doi.org/10.1016/j.jchromb.2018.02.010>.
 20. Citti C, Lagana A, Capriotti AL, Montone CM, Cannazza G. Kratom: the analytical challenge of an emerging herbal drug. *J Chromatogr A.* 2023;1703:464094. <https://doi.org/10.1016/j.chroma.2023.464094>.
 21. de Moraes NV, Moretti RA, Furr EB 3rd, McCurdy CR, Lanchote VL. Determination of mitragynine in rat plasma by LC-MS/MS: application to pharmacokinetics. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2009;877(24):2593–7. <https://doi.org/10.1016/j.jchromb.2009.06.023>.
 22. Hughs M, Kish-Trier E, O'Brien A, McMillin GA. Analysis of mitragynine and speciociliatine in umbilical cord by LC-MS-MS for detecting prenatal exposure to kratom. *J Anal Toxicol.* 2023;46(9):957–64. <https://doi.org/10.1093/jat/bkac064>.
 23. Janchawee B, Keawpradub N, Chittrakarn S, Prasetho S, Waratananurak P, Sawangjareon K. A high-performance liquid chromatographic method for determination of mitragynine in serum and its application to a pharmacokinetic study in rats. *Biomed Chromatogr.* 2007;21(2):176–83. <https://doi.org/10.1002/bmc.731>.
 24. Le D, Goggin MM, Janis GC. Analysis of mitragynine and metabolites in human urine for detecting the use of the psychoactive plant kratom. *J Anal Toxicol.* 2012;36(9):616–25. <https://doi.org/10.1093/jat/bks073>.
 25. Lo Faro AF, Di Trana A, La Maida N, Tagliabracci A, Giorgetti R, Busardo FP. Biomedical analysis of new psychoactive substances (NPS) of natural origin. *J Pharm Biomed Anal.* 2020;179:112945. <https://doi.org/10.1016/j.jpba.2019.112945>.
 26. Meireles V, Rosado T, Barroso M, Soares S, Goncalves J, Luis A, et al. Mitragyna speciosa: clinical, toxicological aspects and analysis in biological and non-biological samples. *Medicines (Basel).* 2019;6(1):1–35. <https://doi.org/10.3390/medicines6010035>.
 27. Neerman MF, Frost RE, Deking J. A drug fatality involving kratom. *J Forensic Sci.* 2013;58(Suppl 1):S278–9. <https://doi.org/10.1111/1556-4029.12009>.
 28. Parthasarathy S, Ramanathan S, Ismail S, Adenan MI, Mansor SM, Murugaiyah V. Determination of mitragynine in plasma with solid-phase extraction and rapid HPLC-UV analysis, and its application to a pharmacokinetic study in rat. *Anal Bioanal Chem.* 2010;397(5):2023–30. <https://doi.org/10.1007/s00216-010-3707-7>.
 29. Wang C, Walker AE. Fatal mitragynine-associated toxicity in Canada: a case report and review of the literature. *Acad Forensic Pathol.* 2018;8(2):340–6. <https://doi.org/10.1177/1925362118782076>.
 30. King TI, Sharma A, Kamble SH, Leon F, Berthold EC, Popa R, et al. Bioanalytical method development and validation of corynantheidine, a kratom alkaloid, using UPLC-MS/MS, and its application to preclinical pharmacokinetic studies. *J Pharm Biomed Anal.* 2020;180:113019. <https://doi.org/10.1016/j.jpba.2019.113019>.
 31. Philipp AA, Wissenbach DK, Weber AA, Zapp J, Maurer HH. Phase I and II metabolites of speciogynine, a diastereomer of the main kratom alkaloid mitragynine, identified in rat and human urine by liquid chromatography coupled to low- and high-resolution linear ion trap mass spectrometry. *J Mass Spectrom.* 2010;45(11):1344–57. <https://doi.org/10.1002/jms.1848>.
 32. Philipp AA, Wissenbach DK, Weber AA, Zapp J, Maurer HH. Metabolism studies of the Kratom alkaloids mitraciliatine and isopaynantheine, diastereomers of the main alkaloids mitragynine and paynantheine, in rat and human urine using liquid chromatography-linear ion trap-mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci.* 2011;879(15–16):1049–55. <https://doi.org/10.1016/j.jchromb.2011.03.005>.
 33. Philipp AA, Wissenbach DK, Weber AA, Zapp J, Maurer HH. Metabolism studies of the kratom alkaloid speciociliatine, a diastereomer of the main alkaloid mitragynine, in rat and human urine using liquid chromatography-linear ion trap mass spectrometry. *Anal Bioanal Chem.* 2011;399(8):2747–53. <https://doi.org/10.1007/s00216-011-4660-9>.
 34. Philipp AA, Wissenbach DK, Weber AA, Zapp J, Zoerntlein SW, Kanogsunthornrat J, et al. Use of liquid chromatography coupled to low- and high-resolution linear ion trap mass spectrometry for studying the metabolism of paynantheine, an alkaloid of the herbal drug Kratom in rat and human urine. *Anal Bioanal Chem.* 2010;396(7):2379–91. <https://doi.org/10.1007/s00216-009-3239-1>.
 35. Philipp AA, Wissenbach DK, Zoerntlein SW, Klein ON, Kanogsunthornrat J, Maurer HH. Studies on the metabolism of mitragynine, the main alkaloid of the herbal drug Kratom, in rat and human urine using liquid chromatography-linear ion trap mass spectrometry. *J Mass Spectrom.* 2009;44(8):1249–61. <https://doi.org/10.1002/jms.1607>.
 36. Tanna RS, Nguyen JT, Hadi DL, Manwill PK, Flores-Bocanegra L, Layton ME, et al. Clinical pharmacokinetic assessment of kratom (*Mitragyna speciosa*), a botanical product with opioid-like effects, in healthy adult participants. *Pharmaceutics.* 2022;14(3):620. <https://doi.org/10.3390/pharmaceutics14030620>.
 37. Avula B, Sagi S, Wang YH, Wang M, Ali Z, Smillie TJ, et al. Identification and characterization of indole and oxindole alkaloids from leaves of *Mitragyna speciosa* korth using liquid chromatography-accurate QToF mass spectrometry. *J AOAC Int.* 2015;98(1):13–21. <https://doi.org/10.5740/jaoacint.14-110>.
 38. CLSI. Liquid-chromatography- mass spectrometry methods. Approved guideline In: C62-A Cd, editor. Clinical and Laboratory Standards Institute. Wayne, PA2014.
 39. Matuszewski BK, Constanzer ML, Chavez-Eng CM. Strategies for the assessment of matrix effect in quantitative bioanalytical methods based on HPLC-MS/MS. *Anal Chem.* 2003;75(13):3019–30. <https://doi.org/10.1021/ac020361s>.
 40. U.S. Department of Health and Human Services FDA. M10 bio-analytical method validation and study sample analysis. Guidance for Industry. November 7, 2022 [Available from: <https://www.fda.gov/media/162903/download>].
 41. Huestis MA, Brett MA, Bothmer J, Atallah R. Human mitragynine and 7-hydroxymitragynine pharmacokinetics after single and multiple daily doses of oral encapsulated dried kratom leaf powder. *Molecules.* 2024;29(5):984. <https://doi.org/10.3390/molecules29050984>.
 42. Berthold EC, Kamble SH, Raju KS, King TI, Popa R, Sharma A, et al. Preclinical pharmacokinetic study of speciociliatine, a kratom alkaloid, in rats using an UPLC-MS/MS method. *J Pharm Biomed Anal.* 2021;194:113778. <https://doi.org/10.1016/j.jpba.2020.113778>.

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